

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Order – (F) C B D A (E) 1 mark for each max 3	3			3		
	(b)			Indicative content (The crystallisation temperature is the temperature at which crystals appear). • {Set up a hot water bath/heat water} and an ice bath. • Put given mass of salt into the boiling tube. • Add given volume water. • Warm the boiling tube in the hot water bath until the salt dissolves. • Put the boiling tube in the ice bath and stir with the thermometer. • Note the temperature at which crystals first appear and record it in the table. • Add 1 cm³ water. • Warm the solution again, stirring until all the crystals dissolve. • Then repeat the cooling and note the new temperature at which crystals appear. • Repeat until 10 cm³ water has been used. Accept alternative methods.	3	3		6		6

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				5-6 marks Comprehensive description of the method above. The water must be warmed until all the salt dissolves. When cooling the temperature of the appearance of crystals is noted. Repeats are also important. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Candidates describe most of the method, e.g. heat water, add salt, add more salt, identify the mass of salt added and change in temperature. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Candidates describe some of the method. E.g. heat the water and add the salt. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit.						

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	(c)	(i)		Sodium sulfate		1		1		1
		(ii)		$115 - 40 = 75\text{g}/100\text{g water} \pm 2\text{ g (73-77 g)}$		1		1	1	1
		(iii)		Any 3 $\times$ (1) from: Sodium sulfate – solubility decreases (1) Potassium nitrate increases (between 30°C and 60°C) (1) Potassium is less soluble than sodium sulfate until 38°C (1) more soluble than sodium sulfate after 38°C (1) solubility is the same at 38°C(1)		3		3	2	4
	(d)			Any 2 $\times$ (1) from: At 10°C the point is on the line (1) 30°C potassium nitrate is 50 (1) 50°C potassium nitrate is 90 - wasn't on the line so no don't agree (1) If no values given (0 marks)			2	2		1
				Question 3 total	6	8	2	16	1	12